configuration information and control commands. Through MQTT, encrypting messages using TLS and verifying clients using modern authentication protocols becomes much easier.

Constrained application protocol (CoAP)

CoAP is a Web transfer protocol for constrained devices and networks in IoT. It can be implemented over a user datagram protocol (UDP), and is designed for applications with limited capacity to connect using light-weight machine-to-machine (LWM2M) communication — such as smart energy and building automation. LWM2M allows remote management of IoT devices, and provides interfaces to securely monitor and regulate them. CoAP architecture is based on the famous REST model, where servers make resources available under a URL, and clients access these resources using methods such as GET, PUT, POST, and DELETE. The CoAP and HTTP protocols have many similarities, with the difference that CoAP is enhanced for IoT, and more specifically for M2M. It has low overhead, along with proxy and caching capabilities, and exchanges messages asynchronously. The architecture of CoAP is divided into two main categories: messaging, which is responsible for the reliability and duplication of messages; and request/ response, which is responsible for communication.

The message layer is on top of the UDP, and is responsible for exchanging messages between the IoT devices and Internet. CoAP has four different types of messages — confirmable, non-confirmable, acknowledgment and reset. A confirmable message (CON) is a reliable message when exchanged between two endpoints. It is sent over and over again until the other end sends an acknowledge message (ACK). The ACK message has the same message ID as that of a CON. If the server faces issues managing the incoming request, it can send back a reset message (RST) instead of an ACK. For exchanging non-critical messages, unreliable NON messages can be used, where the server does not acknowledge the message. NON messages are assigned message IDs to detect duplicate messages.

Request/response is the second layer of the CoAP abstraction layer, which uses CON or NON messages to send requests. In scenarios where a server can respond immediately, a request is sent using a CON message, with an ACK message containing the response, or the error code is sent back by the server. The request and the response both have the same token, different from the message ID. In places where the server cannot respond immediately, it sends an ACK message with no content as the response. Once the response is ready, a new CON message containing the response is sent back to the client and the client acknowledges the response received.

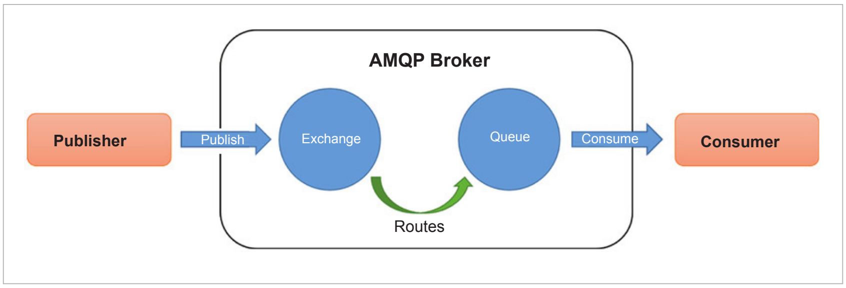

Figure 6: AMQP architecture (Reference: *https://www.researchgate.net/figure/Advanced-message-queueing-protocol-architecture_fig8_305987854*)

Advanced message queuing protocol (AMQP)

AMQP is an open standard application layer protocol designed for higher security, reliability, easy provisioning, and interoperability. It is a connection-oriented protocol, which means the client and the broker need to establish a connection before they transfer data because TCP is used as a transport protocol. AMQP offers two levels of QoS for reliable message delivery — unsettle format (similar to MQTT QoS0) and settle format (similar to MQTT QoS1). The main difference between AMQP and MQTT standards is that in AMQP the broker is divided into two main components — exchange and queues. Exchange is responsible for receiving publisher messages and sending them to queues. Subscribers connect to those queues, which basically represent the topics, and receive the sensory data whenever they are available.

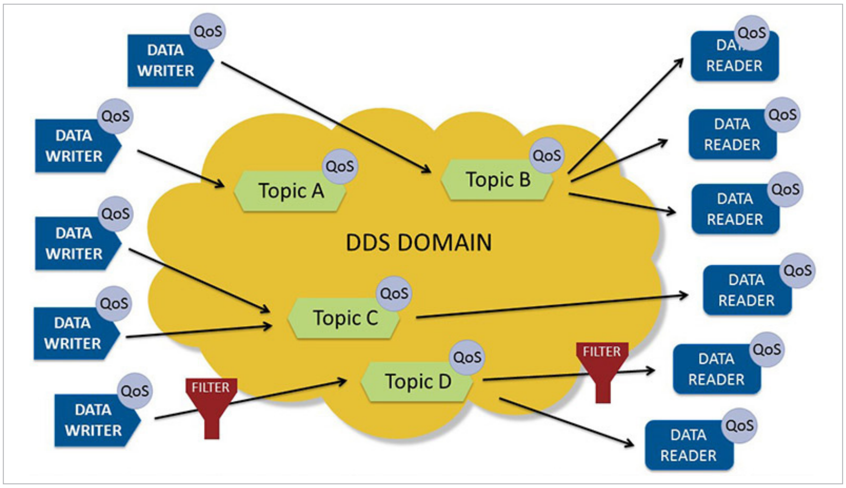

Figure 7: DDS data-centric model (Reference: *https://www.dds-foundation.org/what-is-dds-3/*)